Linear Algebra I: HW5 Partial Solution

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If T is a linear transformation between two vector spaces, we denote $\operatorname{im} T := R(T)$, $\ker T := N(T)$.

5. This is high school algebra.

9. Define $T, U : F^2 \to F^2$ by

$$T(x,y) = (x+y, 0) \quad U(x,y) = (y,0) \text{ for any } (x,y) \in F^2$$

One can check that T, U are linear and that

$$UT(x,y) = U(x+y,0) = (0,0) \text{ for any } (x,y) \in F^2.$$

While

$$TU(x,y) = T(y,0) = (y,0) \text{ for any } (x,y) \in F^2.$$

Therefore, $UT = T_0, TU \neq T_0$.

Let $\beta = \{(1,0),(0,1)\}$ be an ordered basis for F^2, define

$$A = [U]_\beta = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \quad B = [T]_\beta = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$$

Then $AB = [UT]_\beta = O, BA = [AB]_\beta \neq O$.

11. ($\implies$) If $T^2 = T_0$, then for any $y \in \operatorname{im} T$, $y = T(x)$ for some $x \in V$, then $T(y) = T^2(x) = T_0(x) = 0$, $y \in \ker T$. Therefore, $\operatorname{im} T \subseteq \ker T$.

($\impliedby$) Suppose that $\operatorname{im} T \subseteq \ker T$, then for any $x \in V$, $T^2(x) = T(T(x)) = 0$ since $T(x) \in \operatorname{im} T$. Therefore, $T^2 = T_0$.

12. (a) If UT is 1-1, then $\ker UT = \{0\}$. Since $T(x) = 0 \implies UT(x) = 0$, $\ker T \subseteq \ker UT = \{0\}$, thus T is 1-1.

U is NOT necessarily 1-1. For example, define $T : F^2 \to F^3$, $U : F^3 \to F^2$ by

$$T(x,y) = (x,y,0) \quad U(x,y,z) = (x,y) \quad UT = \mathrm{id}_V .$$

Then UT is 1-1 but U is not.

(b) If UT is onto, then $\operatorname{im} UT = Z$,'since if $z \in \operatorname{im} UT$, there exists $x \in V$ such that $z = UT(x)$, put $y = T(x) \in W$, then $z = U(y)$, i.e. $z \in \operatorname{im} U$. That is,

$$Z = \operatorname{im} UT \subseteq \operatorname{im} U \subseteq Z$$

thus, $\operatorname{im} U = Z$, U is onto.

The example above also shows that UT is onto but T is not.

(c) If U, T are 1-1 and onto, for any $x \in \ker UT$, $UT(x) = 0$, since U is 1-1, $T(x) = 0$, but T is also 1-1, $x = 0$, thus $\ker UT \subseteq \{0\}$ and naturally $0 \in \ker UT$. UT is 1-1.

Given $z \in Z$, since U is onto, there exists $y \in W$ such that $z = U(y)$, for this y, there exists $x \in V$ such that $T(x) = y$ since T is also onto. Therefore, $z = UT(x)$, i.e. $Z \subseteq \operatorname{im} UT$ and naturally $\operatorname{im} UT \subseteq Z$.

2. Check that all vector space axioms hold carefully.

3. (a) Since $A(v + w) = Av + Aw, A(av) = a(Av)$ for any $v, w \in F^n, a \in F$, L_A is linear.

(b) Let e_i be the column vector in F^n such that all entries are 0 but the i-th column being 1, and v_j be the column vector in F^m such that all entries are 0 but the j-th column being 1. Then β (resp. γ) is the ordered basis collecting e_i (resp. v_j). Compute

$$L_A(e_i) = Ae_i = \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix} e_i = a_{1i}v_i + \cdots + a_{mi}v_m \text{ for any } i = 1, \cdots, n.$$

Therefore,

$$[L_A]_\beta^\gamma = \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix} = A.$$

4. Write $A = (A_{ij}), B = (B_{ij}), C = (C_{ij})$, then

$$\begin{aligned} (A(BC))_{ij} &= \sum_{k=1}^{n} A_{ik}(BC)_{kj} \\ &= \sum_{k=1}^{n} A_{ik} \left(\sum_{l=1}^{p} B_{kp}C_{pj} \right) \\ &= \sum_{k=1}^{n} \sum_{l=1}^{p} A_{ik}B_{kp}C_{pj} \\ &= \sum_{l=1}^{p} \left(\sum_{k=1}^{n} A_{ik}B_{kp} \right) C_{pj} \\ &= \sum_{l=1}^{p} (AB)_{ip}C_{pj} = ((AB)C)_{ij}, \end{aligned}$$

and

$$\begin{aligned} ((AB)^t)_{ij} &= (AB)_{ji} \\ &= \sum_{k=1}^{n} A_{jk}B_{ki} \\ &= \sum_{k=1}^{n} (B^t)_{ik}(A^t)_{kj} = (B^tA^t)_{ij}. \end{aligned}$$

5. Define $X = \begin{pmatrix} A & B \\ C & D \end{pmatrix} = (X_{ij}), Y = \begin{pmatrix} A' & B' \\ C' & D' \end{pmatrix} = (Y_{ij})$. Then for any $1 \leq i \leq m_1, 1 \leq j \leq p_1$,

$$\begin{aligned}
(XY)_{ij} &= \sum_{k=1}^{n_1+n_2} X_{ik}Y_{kj} \\
&= \sum_{k=1}^{n_1} X_{ik}Y_{kj} + \sum_{k=n_1+1}^{n_1+n_2} X_{ik}Y_{kj} \\
&= \sum_{k=1}^{n_1} A_{ik}A'_{kj} + \sum_{j=1}^{n_2} B_{ik}C'_{kj} \\
&= (AA')_{ij} + (BC')_{ij} = (AA' + BC)_{ij}.
\end{aligned}$$

The similar argument can be apply to prove the other cases.